

MD ENAYET CHOWDHURY

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OBJECTIVE

Seeking an internship in civil engineering, specializing in water resources, to apply expertise in hydrologic modeling, hydro climatology, flood forecasting, machine learning, and GIS-based geospatial analysis.

EDUCATION

Ph.D. in Civil Engineering (in progress; expected graduation: 2029) CGPA: 3.89/4.00 The University of Texas at Austin Austin, Texas, USA Supervised by Dr Jun-Whan Lee	Aug 2024 – Present
M.Sc. in Water Resources Development CGPA: 3.75/4.00 Bangladesh University of Engineering and Technology Dhaka, Bangladesh	Apr 2021 – Sept 2023
B.Sc. in Civil Engineering CGPA: 3.84/4.00 Bangladesh University of Engineering and Technology Dhaka, Bangladesh	Feb 2015 – Apr 2019

TECHNICAL SKILLS

Applications: Delft-SFINCS, Delft-FEWS, ArcGIS, QGIS, AutoCAD, Google Earth Pro, Google Earth Engine, HEC-RAS, HEC-HMS, MATLAB, Microsoft Office Suite | **Coding:** Python, R, XML, C, C++

RESEARCH WORK EXPERIENCES

Graduate Research Assistant The University of Texas at Austin ➤ Analyze the compound flood drivers' individual impact on coastal flooding using a physics-based model, Delft-SFINCS	Aug 2024 – Present
Co-Project Investigator Bangladesh University of Engineering and Technology ➤ Worked on climate modeling, scenarios, and projections, as well as sustainable sediment management for the 22 coastal towns in Bangladesh	Oct 2022 – Aug 2024
Co-Project Investigator Bangladesh University of Engineering and Technology ➤ Developed a component of an open-source flash flood forecasting model, named Delft-FEWS, which integrated meteorological (WRF), hydrological (HEC-HMS), and hydrodynamic models (HEC-RAS) in a single graphical user interface [Publication] ➤ Developed and implemented advanced machine learning models using Python, Google Earth Engine, and RCLimDex, integrating Sentinel-1 SAR data, ArcGIS-based geospatial analysis, and climate indices for flash flood susceptibility mapping [Publication]	Oct 2019 – June 2023
Reviewer Journal of Irrigation and Drainage Engineering – ASCE Convening Organization: American Society of Civil Engineers (Reston, US)	2021

CONSULTANCY EXPERIENCES

Riverbank Survey Specialist Bangladesh University of Engineering and Technology ➤ Quantified the current geotechnical conditions of various coastal embankments/riverbanks using the Standard Penetration Test (SPT) and Electrical Resistivity Meter (ERM) Client: Bangladesh Water Development Board (BWDB)	Jan 2022 – Jan 2023
Junior Remote Sensing Analyst Bangladesh University of Engineering and Technology ➤ Managed a comprehensive water quality data repository for 53 transboundary rivers of Bangladesh using geospatial analysis and mapping tools, including ArcGIS and QGIS Client: The World Bank	Feb 2021 – Jan 2022
Consultant on Hydrological, Hydraulic, and Morphological Study of Rivers Bangladesh University of Engineering and Technology ➤ Worked on data collection through field surveys on the hydrological, hydraulic, and morphological aspects of various rivers in Bangladesh Clients: Local Government Engineering Department (LGED), Bangladesh Water Development Board (BWDB)	Dec 2019 – Nov 2023

TEACHING EXPERIENCES

Assistant Professor (on study leave) Bangladesh University of Engineering and Technology ➤ Mentored a class of 30–40 students, delivering lectures and guiding them in field surveys, as well as conducting Focus Group Discussions (FGDs) and Key Informant Interviews (KIIs).	May 2024 – Present
Lecturer Bangladesh University of Engineering and Technology ➤ Designed and delivered a short course providing hands-on training in climate data downloading, statistical downscaling, and bias correction using R	Oct 2019 – May 2024

- Served as the primary instructor for undergraduate classes of 150–200 students, delivering lectures, arranging course materials and slides, and leading lab sessions across structural, geotechnical, and water resources engineering topics

PUBLICATIONS

Journal Articles (Selected)

1. Chowdhury, M. E., Islam, A. S., Zzaman, R. U., and Khadem, S. (2025). A machine learning-based approach for flash flood susceptibility mapping considering rainfall extremes in the northeast region of Bangladesh. *Advances in Space Research*, Volume 75, Issue 2, Pages 1990-2017. DOI: 10.1016/j.asr.2024.10.047
2. Chowdhury, M. E., Islam, A. S., Lemans, M., Hegnauer, M., Sajib, A. R., Pieu, N. M., Das, M. K., Shadia, N., Haque, A., Roy, B., Billah, M., Abdullah, F., Mamoon, W. B., Kaiser, S., Bala, S. K., Islam, G. M. T., Sarker, G. C., Bhuiyan, S. R., and Bhuyan, A. (2023). An efficient flash flood forecasting system for the un-gaged Meghna basin using open source platform Delft-FEWS. *Environmental Modelling and Software*, 161, 105614. DOI: 10.1016/j.envsoft.2022.105614
3. Islam, A., Islam, M. S., Chowdhury, M. E., and Badhon, F. F. (2021). Influence of vetiver grass (*Chrysopogon zizanioides*) on infiltration and erosion control of hill slopes under simulated extreme rainfall condition in Bangladesh. *Arabian Journal of Geosciences*, 14(2). DOI: 10.1007/s12517-020-06338-y
4. Chowdhury, M. E., Islam, M. S., Alam, T., Barua, S., Shahriar, M. S., and Anisa, H. (2020). Infiltration in vegetated soil: empirical modeling and sensitivity analysis. *Modeling Earth Systems and Environment*, 7(1), 547–559. DOI: 10.1007/s40808-020-00867-x
5. Abdullah, M.S., Nowreen, S., Zzaman, R.U., Hasnat, S., Satu, S.M., Chowdhury, M.E. (2021). Urbanising delta: what lessons Dhaka offers to face challenges. *Journal of Policy & Governance*, 01(02) (December 2021), 1-21. DOI: 10.33002/jpg010201

Conference Proceedings (Selected)

1. Chowdhury, M.E. and Sakib, S. (2021). A review on the pullout capacity of an anchor block embedded in cohesionless soil. *Proceedings of the 5th International Conference on Advances in Civil Engineering (ICACE 2020)*, CUET, 4-6 March 2021, Imam, Rahman and Pal (eds.), Chattogram-4349, Bangladesh.
2. Shampa, Rahman, M.M., Islam, M.S., Rahman, M.A., Chowdhury, M.E., Sakib, M.S., Haque, A., Rahman, M.R., Salehin, M. (2021). Community led approach towards managing river bank erosion. *8th International Conference on Water and Flood Management (ICWFM 2021)*, Institute of Water and Flood Management (IWFM), Dhaka, Bangladesh.

Theses

1. Chowdhury, M.E. (2023). Modelling of basin wide flash flood susceptibility using traditional and hybrid machine learning techniques for the northeast region of Bangladesh. M.Sc. thesis, Bangladesh University of Engineering and Technology (BUET), Dhaka-1000.
2. Chowdhury, M.E. (2019). Effects of vegetation on soil water infiltration. B.Sc. thesis, Bangladesh University of Engineering and Technology (BUET), Dhaka-1000.

AWARDS, ACHIEVEMENTS, AND ADVANCED TRAININGS

Advanced Training Spatial and temporal climate change analysis using CORDEX regional climate models over South Asia for Bangladesh	2021
Dean's Award For maintaining an equal to or greater than 3.75 GPA in one academic year	2016-2019
Best Paper Award 14 th Global Engineering and Technology Conference, 29-30 December 2017, BIAM Foundation, Dhaka, Bangladesh	2017
Honourable Mention (for distinction in co-curricular activities) Notre Dame College, Dhaka	2014

LEADERSHIP EXPERIENCES

Civil, Architectural, and Environmental Engineering Hackathon Mentor The University of Texas at Austin	2025
➤ Guided the participants through challenges, assisted with technical issues related to Python coding, and helped with tasks such as mapping inline in Jupyter Notebook, data visualization, and defining functions in Python	
Assistant Organizing Secretary International Conference on Water and Flood Management (ICWFM)	2023 &
➤ Arranged technical sessions, managed logistics, oversaw print and publication tasks, coordinated with scientific and advisory committee members, and handled venue management	2021
Director Bangladesh University of Engineering and Technology Debating Club	2018-2019
➤ Organized and oversaw university-level debating competitions, managing volunteer teams, selecting debate topics, coordinating with faculty members, and arranging event venues to ensure smooth execution	
Assistant General Secretary (Debate and Workshop) Notre Dame College Debating Club	2014
➤ Arranged weekly debating sessions and organized topic-focused workshops to enhance members' debating skills	
Vice President Ideal School and College Debating Club	2011-12
➤ Led sponsorship negotiations with corporate partners, managed event logistics, and served as editor of the club's annual magazine	